



7988 - Multiplexed Planet Discovery and Binary-Star Disk Demographics in the Young Cluster IC 348 with a Pilot for Full-Frame NIRISS AMI

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	IC348-Fld1	NIRISS Aperture Masking Interferometry	(1) IC348-Fld1

ABSTRACT

Directly imaged planets offer our best window into outer-planet demographics, provide exquisite spectra of planetary atmospheres, and (at young ages) capture planet assembly and circumplanetary disks in action. Binary systems offer repeatable tests of planet-formation models, directly connecting initial conditions (sculpting of disks by stellar companions) to final outcomes (planet demographics in (identical) older binaries).

JWST is revolutionizing both, but it is limited by finite spatial resolution, most acutely for the youngest (but most distant) stellar populations where planets are brightest and planet assembly is ongoing. Ironically, those stars also still saturate. Fortunately, NIRISS AMI (Aperture Mask Interferometry) solves both problems, and the new full-frame mode finally allows multiplexing across the full 2.2 arcmin NIRISS FOV.

We propose an extremely efficient multiplexed pilot survey with NIRISS Full-Frame AMI for the young cluster IC 348, simultaneously observing 25 cluster stars in 4 filters across the NIRISS FOV with just one pointing and costing just 13.2h. We will search for Jovian planets (down to 2 MJup) on solar-system scales (20 AU), testing if dense cluster environments produce as many giant planets as sparse moving groups. We will distinguish

circum-primary/secondary disks within ~10 binaries via resolved colors, assessing where and how quickly planets form in binary systems. Finally, we will optimize strategies needed for future large surveys of planets/disks around hundreds of young stars using multiplexed AMI. Our pilot survey thus achieves cutting-edge stand-alone science, while also unlocking revolutionary future opportunities.

OBSERVING DESCRIPTION

We will obtain NIRISS AMI observations in full-frame mode for one pointing centered just off the middle of the young cluster IC 348. We will expose for a total of 7700-9000 seconds per filter in F227W, F380M, F430M, and F480M, aiming to test the detection limits achievable in each filter and thus splitting the time evenly. We will use 3-5 groups per integration, 7-9 integrations per exposure, 4 primary dithers, and 5 sub-pixel dithers. We do not use additional groups because the brightest stars become nonlinear or even saturate very quickly, but given updated analysis options in the literature, nonlinearity and even moderate saturation now seems to be warranted.

We do not request separate calibrator observations because we aim to test the use of other stars within the same frame to calibrate a given source. This strategy takes advantage of simultaneity, calibrator diversity (location, brightness, color), and larger total number of photons (added over many other sources) to match or exceed the performance of a single mode-matched calibrator at the same detector location as a science target, but observed at a different time. The use of both primary and sub-pixel dithering is also intended to further increase calibrator diversity in terms of sub-pixel centering and location with respect to detector cosmetics. An assessment of the validity of this strategy is a major deliverable for our program, given that it is nearly a requirement to harness the advantages of full-frame AMI.

Proposal 7988 - Targets - Multiplexed Planet Discovery and Binary-Star Disk Demographics in the Young Cluster IC 348 with a Pilot fo...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	IC348-Fld1	RA: 03 44 34.7000 (56.1445833d)	Proper Motion RA: 0.0	
			Dec: +32 07 30.64 (32.12518d)	Proper Motion Dec: 0.0	
			Equinox: J2000	Parallax: 0.0"	
	Comments: Category=Stellar Cluster Description=[Open star clusters, Stellar associations, Young associations, Young star clusters]				

Proposal 7988 - Observation 1 - Multiplexed Planet Discovery and Binary-Star Disk Demographics in the Young Cluster IC 348 with a ...

Observation	Proposal 7988, Observation 1: IC348-Fld1 Tue Jan 27 05:00:25 GMT 2026									
	Diagnostic Status: Warning Observing Template: NIRISS Aperture Masking Interferometry									
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(1)	IC348-Fld1	RA: 03 44 34.7000 (56.1445833d) Dec: +32 07 30.64 (32.12518d) Equinox: J2000		Proper Motion RA: 0.0 Proper Motion Dec: 0.0 Parallax: 0.0"					
	<i>Comments:</i> <i>Category=Stellar Cluster</i> <i>Description=[Open star clusters, Stellar associations, Young associations, Young star clusters]</i>									
Acquisition	#	Target								
	1	NONE								
Template	Subarray									
	FULL									
Dithers	#	Primary Dithers				Subpixel Positions				
	1	4				5				
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	Direct Image
	1	F277W	NISRAPID	3	9	20	180	7730.474		false
	2	F380M	NISRAPID	4	8	20	160	8589.416		false
	3	F430M	NISRAPID	5	7	20	140	9018.887		false
	4	F480M	NISRAPID	5	7	20	140	9018.887		false
PSF References	Additional Justification: true									

Special Requirements	No Parallel Attachments
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